

Problem: Consider a homogeneous, isotropic thin plate of thickness t subjected to a transverse distributed load $p(x, y)$. The coordinate z is measured through the thickness of the plate with the origin located at the mid-plane, such that $-t/2 \leq z \leq t/2$. Assume that the Kirchhoff assumptions of classical plate theory are valid. The three-dimensional equilibrium equations in the absence of body forces are given by

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0, \quad (1a)$$

$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yz}}{\partial z} = 0, \quad (1b)$$

$$\frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0. \quad (1c)$$

The in-plane bending stresses σ_x , σ_y , and τ_{xy} are assumed to be known functions of x , y , and z .

- (a) Using the above equilibrium equations together with the boundary conditions on the top and bottom surfaces of the plate ($z = \pm t/2$), derive the expressions for the transverse shear stresses τ_{xz} , τ_{yz} , and the normal stress σ_z through the plate thickness.
- (b) By integrating the transverse shear stresses through the plate thickness, define the resultant shear forces

$$Q_x = \int_{-t/2}^{t/2} \tau_{xz} dz, \quad (2a)$$

$$Q_y = \int_{-t/2}^{t/2} \tau_{yz} dz, \quad (2b)$$

and use the transverse equilibrium of the plate to obtain the governing differential equation of plate bending in terms of the transverse deflection $w(x, y)$.

Solution:

a) From previous calculations, we have

$$\sigma_x = -\frac{Ez}{1-\nu^2} \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right), \quad (3a)$$

$$\sigma_y = -\frac{Ez}{1-\nu^2} \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right), \quad (3b)$$

$$\tau_{xy} = -\frac{Ez}{1+\nu} \frac{\partial^2 w}{\partial x \partial y}. \quad (3c)$$

By substituting Eqs. (3a) and (3c) into Eq. (1a), τ_{xz} can be obtained

$$\tau_{xz} = -\frac{E}{2(1-\nu^2)} \left(\frac{t^2}{4} - z^2 \right) \left[\frac{\partial}{\partial x} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \right] \quad (4)$$

We know

$$Q_x = -D \frac{\partial}{\partial x} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \rightarrow \frac{Q_x}{D} = -\frac{\partial}{\partial x} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right), \quad D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\tau_{xz} = \frac{3Q_x}{2t} \left[1 - \left(\frac{2z}{t} \right)^2 \right] \quad (5)$$

By substituting Eqs. (3b) and (3c) into Eq. (1b), τ_{yz} can be obtained

$$\tau_{yz} = -\frac{E}{2(1-\nu^2)} \left(\frac{t^2}{4} - z^2 \right) \left[\frac{\partial}{\partial y} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \right] \quad (6)$$

We know

$$Q_y = -D \frac{\partial}{\partial y} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \rightarrow \frac{Q_y}{D} = -\frac{\partial}{\partial y} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right), D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\tau_{yz} = \frac{3Q_y}{2t} \left[1 - \left(\frac{2z}{t} \right)^2 \right] \quad (7)$$

Now, by substituting Eqs. (4) and (6) into Eq. (1c), σ_z can be obtained

$\Rightarrow$

$$\sigma_z = -\frac{E}{2(1-\nu^2)} \left(\frac{t^3}{12} - \frac{t^2 z}{4} + \frac{z^3}{3} \right) \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \right] \quad (8)$$

We know

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) = \nabla^4 w, \nabla^4 w = \frac{p}{D}, D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\sigma_z = -\frac{3p}{4} \left[\frac{2}{3} - \frac{2z}{t} + \frac{1}{3} \left(\frac{2z}{t} \right)^3 \right] \quad (9)$$

b) Substituting Eq. (4) into Eq. (2a),

$$Q_x = -\frac{Et^3}{12(1-\nu^2)} \frac{\partial}{\partial x} (\nabla^2 w) = -D \frac{\partial}{\partial x} (\nabla^2 w) \quad (10)$$

And by substituting Eq. (6) into Eq. (2b),

$$Q_y = -\frac{Et^3}{12(1-\nu^2)} \frac{\partial}{\partial y} (\nabla^2 w) = -D \frac{\partial}{\partial y} (\nabla^2 w) \quad (11)$$

Finally, by writing BC at $z = -t/2$ in Eq. (8),

$$\Rightarrow \nabla^4 w = \frac{p}{D} \quad (12)$$